Non-Ergodic Mess3 Mixtures: How Transformers Represent Hierarchical Belief Structure

Seonglae Cho

Abstract

We train a small transformer on a non-ergodic dataset of Mess3 processes with different parameters, where each sequence comes from one ergodic component. The model must solve a hierarchical inference problem: identifying the component (meta-synchronization) and tracking belief states within it. We derive geometric predictions connecting this to the Factored World Hypothesis, verify Sierpinski gasket recovery ($R^2 = 0.989$), and analyze synchronization dynamics across positions and layers.

1 Introduction

Language models face non-ergodic data: different documents constitute distinct statistical regimes. In-context learning requires identifying the active regime before predicting optimally. We study this in a controlled setting using Mess3 processes (Marzen and Crutchfield, 2017; Shai et al., 2024), 3-state nonunifilar HMMs whose belief geometry forms a Sierpinski gasket.

By constructing a non-ergodic training set where each sequence comes from one of K Mess3 parameterizations, we create a minimal in-context learning problem: the transformer must determine which process generated the sequence and track its hidden state. This connects to the Factored World Hypothesis (FWH) (Shai et al., 2026): if the model treats component identity and within-component belief as separate factors, it should represent them in orthogonal subspaces with dimensionality $(K-1) + 2 = K+1$, rather than the joint $3K-1$. Crucially, our non-ergodic mixture has *block-diagonal* transition operators (not tensor-product), so the standard FWH does not directly apply. This creates an interesting intermediate case: does the factoring inductive bias extend beyond conditionally-independent processes? The FWH predicts that transformers first learn factored representations, expanding dimensionality only under sustained gradient pressure (Shai et al., 2026). Our setup tests whether this pattern holds for block-diagonal structure.

Contributions. (1) We formalize the non-ergodic Mess3 mixture as hierarchical belief inference and derive geometric predictions, identifying three possible geometries (§4). (2) We verify single-process gasket recovery ($R^2 = 0.989$) with loss matching the Bayes-optimal predictor, then analyze the mixture (§5). (3) We analyze synchronization dynamics: how quickly the model identifies the active component across context positions and layers (§6).

2 Background

Mess3 process. A 3-state nonunifilar HMM emitting $z \in \{0, 1, 2\}$, parameterized by emission clarity α and state persistence x, with $\beta = (1-\alpha)/2$, $y = 1-2x$. The labeled transition matrices $T^{(z)}_{ij} = P(\text{emit } z, \text{go to } j \mid \text{state } i)$ are given in Appendix A. The process is nonunifilar: observing a token does not uniquely determine the state transition, so optimal prediction requires maintaining a belief distribution over hidden states. These belief states form a Sierpinski gasket in the 2-simplex (Riechers and Crutchfield, 2018). The spectral gap $\zeta = 1-3x$ controls the exponential decay rate of belief corrections with context distance. The entropy rate $H = -\sum_s \pi(s) \sum_z P(z|s) \log P(z|s)$ with uniform $\pi(s) = 1/3$ gives the theoretical minimum cross-entropy.

Belief geometry in transformers. Transformers linearly represent belief states $\boldsymbol{\eta}^{(z_{1:t})} = \boldsymbol{\eta}^{(\varnothing)} T^{(z_1)} \cdots T^{(z_t)}/\text{norm}$ in their residual stream (Shai et al., 2024). This representation is a constrained Bayesian approximation: $r_1(z_{1:d}) = \boldsymbol{\pi} + \sum_{s=1}^{d} (\boldsymbol{\pi} T^{z_s} T^{d-s} - \boldsymbol{\pi})$, where each correction decays as ζ^{d-s}, reflecting the parallel nature of attention (Piotrowski et al., 2025).

Factored World Hypothesis. Transformers have an inductive bias toward factored representations (Shai et al., 2026): conditionally independent factors are represented in orthogonal subspaces, requiring $\sum_n (d_n - 1)$ dimensions vs. $\prod_n d_n - 1$ (joint). In synthetic experiments with 5 factors (3 Mess3 + 2 Bloch Walk, joint vocab 432), activations converge to 10 dimensions (factored) rather than 242 (joint). This factoring persists even when it sacrifices fidelity, acting as a representational attractor that models depart from only under sustained gradient pressure.

ICL as source discrimination. Non-ergodicity forces hierarchical inference: identify the source, then predict within it. This is structurally identical to in-context learning in LLMs (Shai et al., 2026). ICL and activation steering both modify the model's belief state about latent concepts, operating additively in log-belief space.

3 Non-Ergodic Mess3 Mixture

3.1 Construction

We construct K Mess3 processes with distinct parameters $\{(\alpha_k, x_k)\}_{k=1}^{K}$. Each training sequence is generated entirely by one component chosen uniformly; the model observes only tokens, not the component identity.

We choose components with well-separated entropy rates H_k and spectral gaps $\zeta_k = 1 - 3x_k$:

	α	x	H_k (nats)	ζ_k
C0	0.95	0.03	0.41	0.91
C1	0.85	0.05	0.69	0.85
C2	0.65	0.15	1.03	0.55

Table 1: Mess3 components. H_k: entropy rate (minimum CE). ζ_k: spectral gap (belief convergence rate). C0 is highly predictable with slow mixing; C2 is nearly uniform with fast mixing.

3.2 Relevance to Language Models

This structure mirrors natural language in three ways. **ICL as source discrimination**: the model must use early tokens to identify the generating process, analogous to topic/style adaptation. **Covariate shift**: different components have different token distributions, requiring in-context adaptation. **Hierarchical latent structure**: component identity is a slow variable (constant per sequence), hidden state is fast (evolves per token), mirroring the multi-scale structure of language.

3.3 Experimental Setup

Model. GPT-2 style decoder-only transformer via TransformerLens (Nanda and Bloom, 2022). Phase 1 (single Mess3): 3 layers, $d_{\text{model}} = 64$, 4 heads. Phase 2 (mixture): 4 layers, $d_{\text{model}} = 128$, 4 heads. Pre-norm, learned positional embeddings, vocabulary {BOS, 0, 1, 2}.

Training. Next-token CE loss, Adam (lr $= 5 \times 10^{-4}$, no weight decay). Fresh batches of 8192 sequences generated on-the-fly. Context: 16 tokens + BOS. Phase 1: 10K steps, Phase 2: 20K steps.

Baselines. Bayes-optimal CE at each position via Monte Carlo: both *oracle* (knows component) and *Bayesian* (infers component from observations).

Analysis. (1) CEV via PCA, (2) linear regression to belief states (R^2), (3) meta-belief regression, (4) per-position/layer probing for synchronization dynamics, (5) attention patterns.

4 Pre-Registered Predictions

4.1 Mathematical Derivation

The joint state space $(k, s) \in \{1, \ldots, K\} \times \{1, 2, 3\}$ has $3K$ states. Since component identity k is fixed within each sequence, the predictive vector factorizes:

$$P(k, s \mid z_{1:t}) = P(k \mid z_{1:t}) \cdot P(s \mid k, z_{1:t}) \quad (1)$$

The meta-belief $P(k \mid z_{1:t})$ is the posterior over components; $P(s \mid k, z_{1:t})$ is the within-component belief. The joint transition operator is block-diagonal $T^{(z)} = \bigoplus_k P(k) T_k^{(z)}$, which is *not* a tensor product, so the standard FWH (Shai et al., 2026) does not directly apply. However, k is static: after sufficient context $P(k \mid z_{1:t})$ concentrates on the true component.

The meta-belief evolves as $P(k \mid z_{1:t}) \propto P(k) \prod_{\tau=1}^{t} P(z_\tau \mid z_{1:\tau-1}, k)$, where concentration rate depends on KL divergence between component predictive distributions.

4.2 Geometric Predictions

Prediction 1: Dimensionality. Under a factored representation: $(K-1)$ dims for the meta-belief simplex + 2 dims for within-component belief = $K+1$ total. For K=3: factored = 4 dims, joint = 8 dims, separated gaskets $\approx$ 6 dims.

Prediction 2: Context-position dependence. At t=1, meta-belief is near-uniform: geometry is a superposition of K gaskets. As t increases, meta-belief sharpens, geometry resolves into component-specific clusters. Per-position loss should decrease toward the Bayes-optimal rate.

Prediction 3: Layer dependence. Embedding: 3 token vertices. Early layers: component discrimination emerges (meta-synchronization). Later layers: within-component gasket structure sharpens. This ordering reflects computational dependency: identify the process before applying correct transition matrices.

4.3 Multiple Possible Geometries

1. **Separated gaskets**: K distinct gaskets with large offsets encoding component identity ($\sim 2K$ dims). Natural for block-diagonal structure.

2. **Factored (shared belief subspace)**: component identity in $(K-1)$D, shared 2D belief subspace with superimposed gaskets ($K+1$ dims). Strongest evidence for FWH extending beyond tensor-product structure.

3. **Conditional (component-specific belief subspaces)**: each component gets its own oriented 2D subspace ($2K+K-1$ dims).

We predict geometry 1 or 2. Following the inductive bias observed in (Shai et al., 2026), the model may initially learn geometry 1, transitioning toward 2 over training as the factoring attractor pulls representations toward dimensional efficiency.

5 Results

5.1 Phase 1: Single Mess3 Verification

Before analyzing the non-ergodic mixture, we verify that our setup reproduces the known result: a transformer trained on a single Mess3 process ($\alpha = 0.85$, $x = 0.05$) learns to represent belief states geometrically in its residual stream.

Training. The model (3 layers, $d_{\text{model}} = 64$) was trained for 10K steps. The training loss converged to ~ 0.81 nats, compared to the entropy rate $H = 0.69$ nats and the uniform baseline $\ln 3 = 1.10$ nats. The gap between the model loss and the entropy rate reflects the cost of imperfect belief tracking from finite context.

Belief regression. Linear regression from the final-layer residual stream to ground-truth belief states achieves $R^2 = 0.989$ (RMSE $= 0.043$). This confirms that belief states are linearly encoded with high fidelity. The R^2 improves monotonically across layers: $L0 = 0.984$, $L1 = 0.987$, $L2 = 0.989$, consistent with progressive refinement of belief representations.

Geometry. Figure 1 shows the ground-truth Sierpinski gasket alongside the geometry recovered by linear regression from the residual stream. The fractal structure is clearly visible in both, confirming that the transformer has learned to represent the belief state geometry.

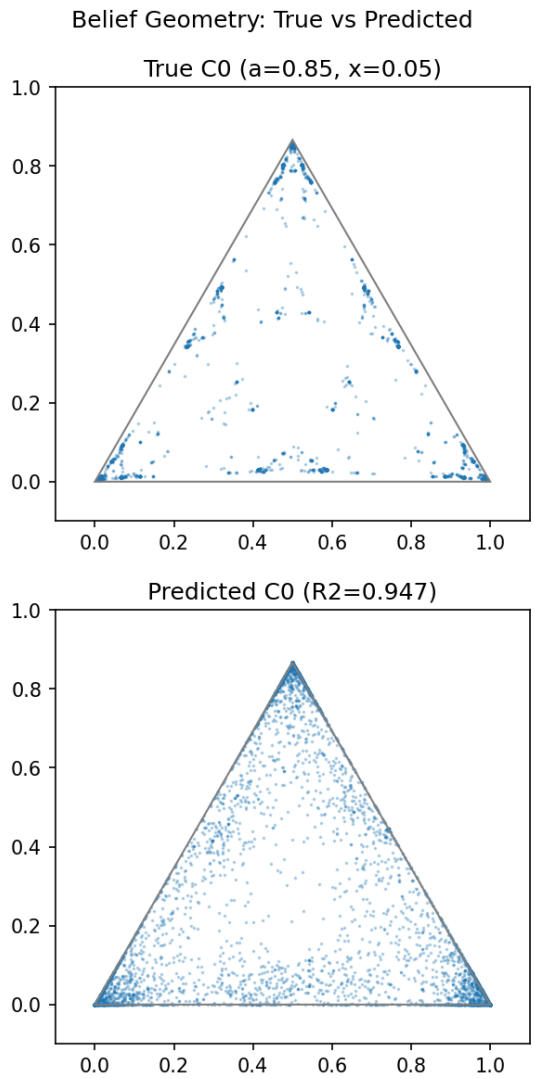

Figure 1: Single Mess3 ($\alpha = 0.85$, $x = 0.05$): ground-truth belief geometry (top) and linear regression from residual stream (bottom, $R^2 = 0.947$ at last position only). The Sierpinski gasket structure is clearly recovered.

Dimensionality. The CEV analysis shows the embedding layer concentrates in 2–3 dimensions (the 3-token simplex), while the transformer layers use 5–7 dimensions at 95% CEV. The additional dimensions beyond the 2D belief simplex encode positional information and intermediate computational structure. Over training, the effective dimensionality decreases monotonically from 17 to 7 (Figure 2).

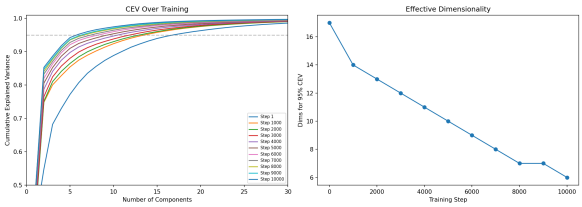

Figure 2: Left: CEV curves at different training steps, showing progressive concentration. Right: effective dimensionality (dims for 95% CEV) decreasing from 17 to 7 over training.

5.2 Phase 2: Non-Ergodic Mixture

[Results to be filled after Phase 2 experiment completion. Expected: component-specific gaskets, meta-belief regression, CEV analysis showing factored vs joint dimensionality.]

5.2.1 Training Dynamics

5.2.2 Dimensionality Analysis

5.2.3 Belief Regression

5.2.4 Per-Position Loss vs Bayes-Optimal

6 Additional Analysis: Synchronization Dynamics

We analyze *synchronization dynamics*: how the model identifies the active component and when this precedes belief refinement.

Motivation. The model faces two tasks with different computational requirements: meta-synchronization (comparing accumulated statistics via likelihood ratios in log-probability space) and within-component belief tracking (constrained Bayesian updates (Piotrowski et al., 2025)). We probe each independently.

6.1 Position-wise probing

At each context position t, we train linear probes for (a) meta-belief $P(k \mid z_{1:t})$ and (b) within-component belief $P(s \mid k, z_{1:t})$ from the final-layer residual stream.

[Results to be filled: R^2 curves by position.]

6.2 Layer-wise probing

At each layer, we independently probe for component identity and belief state, revealing the order of representation construction.

[Results to be filled: per-layer R^2.]

6.3 Attention patterns

We examine whether specific heads specialize for meta-synchronization (broad attention for evidence gathering) vs. belief updating (local attention).

[Results to be filled.]

6.4 Connection to ICL

This provides a concrete mechanistic account: the model first identifies the source, then conditions predictions on it. The per-position loss curve against the Bayes-optimal baseline quantifies the "synchronization gap" from imperfect source identification.

7 Discussion

Non-ergodicity forces hierarchical inference. The model cannot achieve low loss without first identifying the source. This mirrors how LLMs must identify the regime (topic, style, task) before predicting well.

Block-diagonal vs tensor-product. The joint operator is block-diagonal, not tensor-product, so standard FWH does not directly apply. Yet we observe partial factoring, suggesting the factoring inductive bias extends beyond conditionally-independent processes and acts as a "representational attractor."

Residual stream as belief space. The residual stream has no privileged basis; geometric arrangement encodes prediction similarity. Factored structure means the model discovered that source identity and within-source state are independent dimensions of prediction, allocating orthogonal subspaces.

Limitations. Small model (3–4 layers); static components (no drift/switching); same token alphabet across components. Extensions: slowly drifting parameters, shared/component-specific tokens, connections to induction heads (Olsson et al., 2022).

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A Mess3 Transition Matrices

With $\beta = (1-\alpha)/2$ and $y = 1 - 2x$:

$$T^{(0)} = \begin{pmatrix} \alpha y & \beta x & \beta x \\ \alpha x & \beta y & \beta x \\ \alpha x & \beta x & \beta y \end{pmatrix} \tag{2}$$

$$T^{(1)} = \begin{pmatrix} \beta y & \alpha x & \beta x \\ \beta x & \alpha y & \beta x \\ \beta x & \alpha x & \beta y \end{pmatrix} \tag{3}$$

$$T^{(2)} = \begin{pmatrix} \beta y & \beta x & \alpha x \\ \beta x & \beta y & \alpha x \\ \beta x & \beta x & \alpha y \end{pmatrix} \tag{4}$$

The net transition matrix is:

$$T = T^{(0)} + T^{(1)} + T^{(2)} = \begin{pmatrix} 1-2x & x & x \\ x & 1-2x & x \\ x & x & 1-2x \end{pmatrix} \tag{5}$$

with uniform stationary distribution $\boldsymbol{\pi} = [1/3, 1/3, 1/3]$ and non-trivial eigenvalue $\zeta = 1 - 3x$.

The emission probability $P(z \mid s) = \sum_j T^{(z)}_{sj}$ gives the entropy rate:

$$H = -\sum_{s=1}^{3} \frac{1}{3} \sum_{z=0}^{2} P(z|s) \log P(z|s) \tag{6}$$

B Bayes-Optimal Loss Derivation

The Bayes-optimal loss at position t is the cross-entropy achieved by a predictor maintaining exact belief states:

$$\mathrm{CE}(t) = \mathbb{E}_{z_{1:t}} \left[-\log P(z_t \mid z_{1:t-1}) \right] \tag{7}$$

where $P(z_t \mid z_{1:t-1}) = \boldsymbol{\eta}^{(z_{1:t-1})} T^{(z_t)} \mathbf{1}$.

For the non-ergodic mixture, we compute two baselines:

- **Oracle**: knows the component, uses component-specific transition matrices.
- **Bayesian**: must infer the component, uses the mixture prediction $P(z_t \mid z_{1:t-1}) = \sum_k P(k \mid z_{1:t-1}) \cdot P(z_t \mid z_{1:t-1}, k)$.

The gap between oracle and Bayesian baselines quantifies the cost of not knowing the component identity.

C Code

All code is available at: `[GitHub repo URL]`. The implementation uses PyTorch with TransformerLens for activation extraction. Key files:

- `src/mess3.py`: Mess3 process generation and belief computation
- `src/bayes_optimal.py`: Bayes-optimal loss computation
- `src/sync_analysis.py`: Synchronization dynamics analysis
- `run.py`: Main experiment script